IMPROVE LOGISTICS OPERATIONS

Certification Planner LLC

Proprietary and Confidential

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Abstract

This case provides an insight into the experience of redesigning in-plant logistics operations of a large manufacturing complex in India dealing in polymers and polyesters. The complex under reference is expected to have a traffic volume of one truck per one minute by 2015.

The existing systems and procedures to receive and inspect a truck, load material, and complete commercial formalities are designed with a target truck turnaround time of 82 minutes. However, in reality the actual truck turnaround time is significantly higher than the target turnaround time.

The plant is located in a growing industrial area. Inordinate delay in truck turnaround time is a major de-motivating factor for the truck companies to place their trucks with this complex. Consequently, there is a significant variation between the planned and actual dispatch of finished goods.

In order to improve the turnaround time substantially, there is a need to redefine work, and fundamentally change the underlying process. Accordingly, several initiatives are identified to ensure dramatic reduction in the truck turnaround time.

1. The Plant, Location, and Products

Supreme Polymers Ltd. (SPL), was characterized by aggressive growth, (revenue and profitability), dominant production capacity, proven financial and operational performance, and excellent relationship with shareholders. The integrated manufacturing facility of SPL, processing a variety of products, broadly categorized as polymers and polyesters, was located in the western part of India.

The SPL plant location had its own advantages and disadvantages related to logistics. It had easy access to sea route, but trucks were not easily available. The major transport services were provided and operated from Mumbai with a small office at the plant site.

The sales tax advantage applicable to the area had attracted large industrial houses to set up their manufacturing and processing facilities around the SPL plant. This led to greater demand for trucks to move material. The end customers (both small and medium) of SPL were scattered all over the country. To get the sales tax advantage SPL had to use the plant location as a base either for direct invoicing or stock transfer.

The logistics function involved a large number of agencies within and outside SPL, viz. production, quality control, bagging, warehouse, excise, commercial, load office, weighbridge, security, Market Operations Group (MOG) at Mumbai and at plant location, business planning, transport contractors (at Mumbai and at plant location) and transport representatives. Roughly two thirds of the logistics related workforce of SPL was on contract.

2. Systems, Procedures, and Practices

Production Planning: Based on the market forecast and the existing stock at various locations and regions a monthly product wise production plan was finalized at Mumbai and communicated to the plant. The manufacturing department translated this into weekly and daily production plans. There were however, significant variations between what was originally planned and the actual production.

Subject to technical quality clearance, the material produced was stored in a silo for subsequent bagging. The major interface between production and in-plant logistics was the bagging operation, which was under the supervision of Production. The bagging plant was highly automated and bagged material in 25 Kg. standard bag sizes. Along with bagging, the lot number, grade, bag-number, and date were printed for subsequent product, grade, and lot identification. Manual registers kept track of material bagged. Automatic weighing counters cross checked the quantities bagged. The most important function of the bagging section, from the logistics perspective, was related to material accounting. This was the first location in the plant where weight based proper material accounting was possible.

After bagging, material equivalent to 1 Ton was palletized and moved on to an appropriate location, which was determined by the bagging supervisor. The computer stock was updated at the end of the shift on warehouse location occupancy and stock quantity. The forklift operators, who moved material from bagging plant to storage, were under contract to SPL.

Marketing Interface: SPL-Marketing set up was at Mumbai. The marketing executives usually reported to their respective business heads. Based on the market dynamics, dispatch orders were placed on plant location for material delivery. The Mumbai office electronically communicated the set of Dispatch Orders (DO) that were to be processed on any day to plant and a backup confirmation copy was sent by post. This document initiated activities related to material dispatch from SPL complex. The DO copy was received by Marketing Operations Group (MOG) at plant for further processing. Meanwhile, marketing informed the truck operator's representative at Mumbai for placement of trucks in the plant.

The truck operators at Mumbai, in turn, informed their counterparts at the plant for trucks placement. This activity happened in parallel to the transmission of DOs. The MOG at SPL also contacted the truck representatives at plant location for deployment of trucks.

Documentation: After reporting for loading, the trucks were parked in SPL complex for preparation of entry ticket. The transport representative periodically, on arrival of 2 or 3 trucks, visited the dispatch office (MOG location) to verify whether there was any Dispatch Order (DO) to be executed by his company. If yes, he would get a copy of the DO from MOG and prepare the Transport Authority Letter (TAL), a legal document required for material transport. The truck driver along with the DO and TAL presented the truck for security pre-inspection. The pre-inspection was executed by security in batches of approximately five trucks. On security clearance, the trucks were allowed to go inside the SPL manufacturing complex with an entry ticket, a prime document that tracks the truck movement inside the complex.

The entry ticket had a unique number representing the date and the serial number of the truck for the day. The entry ticket had information on the product, customer identification, customer location, and data on vehicle identification. The entry ticket stayed with the vehicle until the truck left the SPL complex. While some of the data elements related to the entry ticket were captured online in the existing computer network, most of the data elements were entered manually.

After generation of the entry ticket, the truck was allowed to pass through security gate and was inspected at the check post for safety features and possible additional material, which might distort the weight of the empty truck. On clearance by Security at the check post, the truck passed through the weighbridge where initial weight of the truck had been recorded.

After the weighbridge entry operation, the truck along with the DO reached the warehouse of the appropriate product group. The truck driver produced the entry ticket, along with DO in the dispatch office. Computer generated blank pick-up notes were provided to the supervisor in charge of warehouse operations to initiate and monitor material loading. A group of contract workers and forklift operators transferred material from a particular location in the warehouse to the assigned truck. While loading was in progress documentation related to lot number, grade and serial number of the bags were entered in the pick-up note. On completion of loading operation, information on bag-wise quantity loaded in the truck was sent to the dispatch office in the form of filled-in pick-up slip for invoice preparation, material accounting, stock updating and other legal formalities. Meanwhile, the truck moved out of the loading bay and was attended to for related activities like tarpaulin tying.

Once the dispatch office in the warehouse received the loading information related to bags, an invoice was generated. Technically, this was to be prepared online. But, in reality, this invoice generation was a batch process. The invoice generation process waited until 4 or 5 trucks were loaded and the related information on pick-up notes was received.

Subsequently, the pick-up note data entry was made based on which the invoices and excise gate passes were generated. The truck driver collected this invoice and moved to the outweigh bridge. Stock and inventory were updated at the end of the day.

When the truck left the warehouse area it was expected to report to the out-weighbridge as early as possible. The truck driver again reproduced his entry ticket at the weighbridge, the truck got weighed, and the material weight (as a difference of two weighbridge recordings) was computed (manually) by the security guard. If the difference suggested a potential problem area, further probing and checking were conducted by security. Otherwise, the truck was allowed to pass the out-weighbridge (See the operations diagram for additional information and clarity).

The Truck Drivers: The following is a set of eight occasions when the truck driver not only needed to interact with an agency in SPL, but also awaited either for a document generation or updating.

- Reporting to MOG at the plant site via truck operators' representative

- Presentation of the truck for security verification before entry ticket generation
- Presentation of the truck for security verification (check- post) after entry ticket generation
- Weighbridge-in entry
- Arrival of the truck at the warehouse for material loading.
- Preparations of the truck like tarpaulin tying etc. after loading.
- Presentation of the truck on the return path at the weighbridge out gate.
- Presentation of documents before exit from SPL.

Security and Logistics Interface: In the SPL complex, the role of security was comprehensive. Their responsibility spanned from material accounting, protection of the complex from fire and accidents, regular supervisory control on pilferage, control of man and material movement and traffic management. Roughly two thirds of the security force was on contract. The orientation and approach by security staff in SPL, in dealing with truck operators was more biased towards control and prevention of material pilferage rather than traffic management.

3. Performance Measures

Truck Turnaround Time: The In-plant logistics system that was practiced at SPL was influenced by the organization of facilities like security gate, weighbridge, layout of loading bay and warehouse occupancy, and procedures followed within SPL. The understanding and co-operation of truck operators and contractors in doing business with SPL also played an important role in adopting and streamlining this process. The result of all these components was the time a truck spent in the system (to be received, loaded and moved out of the system). This single measure not only evaluated the total performance of in-plant logistics function at SPL but also provided an opportunity to quantify performance improvement as and when they were planned and implemented. In order to appreciate this measurement, SPL collected data on truck arrival and departure statistics for a span of 45 days and conducted some analysis. There was no specific reason for selecting this time period and this period was considered to be representative. This data was organized as follows:

- Entry gate time to warehouse reporting time;
- Loading time at the warehouse;
- Exit gate time from entry gate reporting time.

In addition, data was classified as the total cycle time and non-loading time to reflect the time spent by the truck in the system after loading (Table 6, 6A, and 6B).

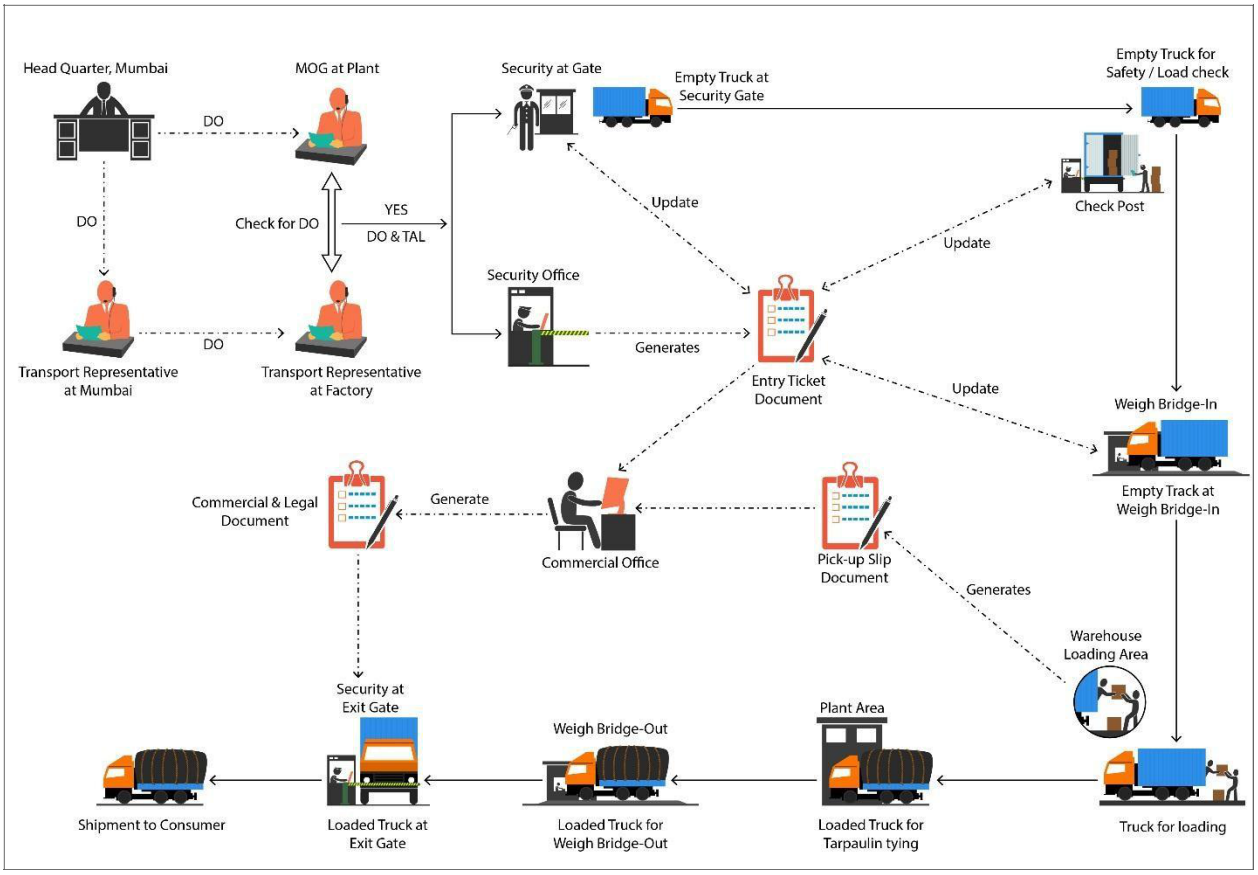


Table 1 – Traffic Load at Plant

Product	# Trucks per day	
	Average	Peak
1. Polymers		
Product 1	100	160
Product 2	162	243
Product 3	139	201
Packing Material	30	45
Waste	5	5
Total Polymers	436	654
2. Polyster		
Product 1	46	70
Product 2	73	125
Product 3	37	49
Packing Material	25	30
Waste	5	5
Total Polyster	186	279
Grand Total	**622**	**933**

Table 2 – Transit times

Summary Statistics: Cycle Time

Total Transit Time (minutes)

	Product 1	Product 2
AVG	164.18	154.52
Min	95.63	92.52
Max	511.89	331.19
Median	144.4	145.39

Additional Details for Product 1

	Loading time	Other than warehouse time	Entry gate to warehouse time
AVG	26.42	137.77	43.36
Min	11.44	77.12	15.89
Max	102.39	409.51	153.92
Median	21.71	126.26	38.76

Additional Details for Product 2

	Loading time	Other than warehouse time	Entry gate to warehouse time
AVG	26.26	128.16	38.36
Min	14.42	61.88	15.33
Max	59.49	314.76	121.36
Median	22.14	123.74	33.52

Exercises and Time Estimates

Case study reading by the participants (30 min)

Exercise 1 (1 hr)

Project Charter

The management team is keen on starting improvement initiative for reducing the truck turn-around time. Management team has appointed a team to use six sigma methodologies to improve the in-plant logistics operations. Refer the case study on SPL to create Project Charter covering:

- Business case
- Problem Statement
- Goal Statement
- Scope
- Timelines
- Key Roles

Exercise 2 (30 minutes)

Calculating DPMO and Sigma Value (Z)

In the table below data for the Truck Turn Around time for Product 1 for 30 days is given. Using the data, calculate mean, median, mode, DPMO, and Sigma value for the logistics operations process capability. The target value for performance metric, Truck Turn-Around Time is 82 minutes.

Day	Turn Around Time (Minutes)
1	153
2	98
3	97
4	156
5	178
6	150
7	81
8	111
9	120
10	198
11	143
12	110
13	137
14	190
15	201
16	95
17	80
18	180
19	94
20	143
21	169
22	99
23	512
24	318
25	256
26	79
27	173
28	211
29	109
30	271

Exercise 3 (1 hr)

Cause and Effect Analysis

Identify the key problem in In-Plant Logistics operations at SPL and conduct cause and effect analysis to identify possible causes.

Exercise 4 (1 hr)

Create QFD for re-engineering the In-plant logistics operation for SPL.

Create a QFD for the best solution identified for improving the In-plant Logistics operations at SPL.

Identify the key elements and the relationships between customer requirements and correlation between various process characteristics.

***The exercise duration includes working on the exercise and 15 minutes of presentation by participants.